

EDUCATION

Columbia University, New York, NY Sep '22 - May '26
BA, Computer Science — GPA: 3.80/4.00 — D1 Varsity Lightweight Crew — Dean's List

TECHNICAL SKILLS

- **Advanced:** Python, Java, JavaScript. **Intermediate:** TypeScript, SQL, C(++), HTML/CSS. **Proficient:** R, $\LaTeX$.
- *Relevant Coursework:* Machine Learning, Artificial Intelligence, Natural Language Processing, Large Language Models, Analysis of Algorithms, Computer Networks, Cloud Computing (AWS), Ethics of LLMs
- *ML/AI:* PyTorch, Transformers, LoRA/QLoRA Fine-Tuning, LLM Pipelines, Pydantic, NumPy, Pandas, Jupyter
- *Infrastructure:* FastAPI, Flask, AWS (Lambda, S3, DynamoDB), MongoDB, Docker, Git, CI/CD, REST APIs

PROFESSIONAL EXPERIENCE

Columbia University Computer Science, *Teaching Assistant*, New York, NY Sep '25 - May '26

- Led weekly office hours for **200+ students** in theory and applications for **Cloud Computing**.
- Code reviewed and graded **50+ student projects** with **AWS management** and **microservice deployment**.

Garmin Ltd., *Software Engineer Intern, Device Interactions*, Olathe, KS May '25 - Aug '25

- Shipped new failover system for **public Garmin Weather API**, improving endpoint error rate by **99%**. Leveraged Java Spring Boot and **Redis state management**.
- Designed backend service for wearables (**deployed Q1 2026**) with comprehensive **error handling** and **unit testing**.
- Worked within **agile methodologies (CI/CD)** to **deploy code to production**; presented demos at weekly stand-ups.

Covet Inc., *Full-Stack Software Engineer Intern*, Seattle, WA May '24 - Aug '24

- Launched **company-first AWS-based onboarding service** that facilitated **first 100+ new client acquisitions**.
- Implemented **test-driven error detection** using Jest and created a sitewide **Rule Engine**, reducing client configuration errors by **99%+**.
- Built **document generation service** with deterministic validation to streamline legal processes.

University of Chicago Dept. of Economics, *Prof. John List Research Assistant*, Chicago, IL May '23 - Aug '23

- Won **department-wide intern research competition** by writing **data-driven** paper on brain-drain.
- Reduced citation time by **80%** for *The Voltage Effect* (ISBN: 9780226820675) with **automated DB API services**

PROJECT EXPERIENCE

Polarize — AI-powered endurance training platform with fine-tuned LLM coach Personal Project

- Fine-tuned **LoRA adapter** (rank-16, **QLoRA 4-bit quantization**) on **Mistral-7B** using **supervised fine-tuning (SFT)** for directed coaching with **structured JSON + .FIT file output** enforcement.
- Built **multi-step agentic pipeline**: context aggregation → LLM inference → Pydantic validation → database commit with dry-run preview and **error recovery**.
- Developed **numeric extraction pipelines** for FIT files with rolling power calculations, outlier filtering, and **deterministic guardrails** for production reliability.

Erg Room Tracker — RFID-based attendance system for Columbia Lightweight Rowing Personal Project

- Built **real-time presence tracking** system serving **30+ D1 athletes** using Raspberry Pi, RC522 RFID, and **Flask/HTMX**. Deployed via **Cloudflare Tunnel**.

Pokébank — Blockchain-based trading platform Computer Networks

- Built decentralized **P2P trading system** with **Merkle tree** hashing, fork resolution, and **nonce mining** for tamper-proof transactions.

BillsWithFriends — Cloud-based Budgeting SaaS Cloud Computing

- Developed a receipt parsing and group-budgeting platform with fully **documented APIs** and session handling, deployed in a **scalable architecture** using various **AWS microservices** and compute strategies.

LEADERSHIP & ACHIEVEMENTS

IRA National Championship, *Bronze Medalist* May '25

- Achieved **bronze medal** at IRA National Championship as member of Columbia Men's Varsity Crew team.
- Maintained **4.0+ GPA** while managing **40+ hr/week** athletic commitment.

Washington University St. Louis Olin Big IdeaBounce, *Runner-Up Venture Award with Cash Prize*
Eagle Scout with Silver Palm, *Boy Scouts of America*